Angelo Tutoring Platform

Project Status Report and Decisions Required

18 September 2026

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Prepared for	[Client name]
Prepared by	[Your company]
Version	1.0 — for discussion

1. Purpose of this report

This report sets out the current state of the Angelo Tutoring application, the work completed in the past week, the infrastructure we propose for launch and its running cost, and the decisions we need from you to plan the remaining work. It is based on a full review of the source code carried out on 17 and 18 September 2026, not on a feature list.

Percentages in this report are engineering estimates of effort remaining. They are not delivery dates. A timeline can be produced once the decisions in Section 6 are made.

2. Executive summary

The application's software foundation is approximately 70% complete. The curriculum content is approximately 15% complete. Taken together, and allowing for the fact that students value content more than infrastructure, the product is roughly 45 to 50% of the way to a version that could be offered commercially.

Measure	Position at 18 September 2026
Software foundation	Approximately 70% complete
Curriculum content	Approximately 15% complete
Product overall	Approximately 45 to 50% of a commercially ready version
Topics with verified videos loaded	28 of 28 (English); 26 of 28 (Afrikaans)
Practice questions in the bank	6 (one topic)
Automated tests	388 passing on every change
Deployed environment	None yet

Three points deserve your attention before anything else:

1. **Content strategy.** The platform is now built to play curated YouTube lessons inside the app rather than to host lessons written from scratch. This week we identified, verified and loaded provisional video selections for all 28 curriculum topics in English, and for 26 of them in Afrikaans. These selections are a starting point for your review, not a final choice.
2. **Security.** Our review found that the application's server routes did not verify the identity of the caller. This has been corrected this week. The correction passed two rounds of independent security review and has been merged. No student data was exposed because the platform has not yet been deployed with real users.

3. **Decisions.** Six decisions listed in Section 6 change the scope of the remaining work substantially. We recommend these are answered first. The remaining questions can follow at your convenience.

3. Current state of the application

3.1 Summary by area

Area	Complete	Comment
Application foundation (login, navigation, mobile install, English and Afrikaans interface)	80%	Password reset is not yet built.
AI Skills course (five modules, prompt challenge, certificate)	85%	The most complete feature in the product.
AI tutor “Angelo” (hints, wrong-answer help, photo scan)	65%	Built but not yet tested against a live AI account.
Administration panel (topics and questions)	65%	Cannot edit lesson text or video selections.
Past papers (50 NSC papers, 2019 to 2024)	55%	Screen shows a fixed list; papers are hosted on a third-party site.
Quiz, points, streaks and progress tracking	40%	Screens exist; results are not yet saved.
Hosting and release pipeline	40%	Automated checks are now in place; nothing is deployed.
Curriculum content (curated video)	15%	Player and structure built; provisional videos loaded this week.
Premium tier and payments	0%	No code exists.
Email	0%	No code exists.

3.2 What is working

Feature	Status	Detail
Sign up and log in	Built	Email and password. Protected pages redirect to the login screen. No Google or Apple sign-in and no password reset yet.
English and Afrikaans interface	Built	84 interface strings translated. The AI tutor also answers in Afrikaans. Lesson content itself is English only.
Install as an app	Built	Installs on phones from the browser and caches topic pages and the formula sheet for offline reading.
AI Skills course	Built	Five modules, a prompt-writing challenge graded by AI, and a downloadable certificate with a public verification link and QR code.
Formula sheet	Built	Static reference, available offline.
In-app video player	Built	YouTube videos play inside the topic page. Each video slot can carry a separate Afrikaans video. The link that previously sent students out to YouTube has been removed.

Feature	Status	Detail
AI tutor (Angelo)	Built, untested live	Six modes and a defined tutoring personality. Daily limits for free users (five hints and one photo scan). Not yet exercised against a live AI account.
Administration panel	Built, untested live	Create topics, write multiple-choice questions with worked solutions, and generate solution steps with AI. Cannot edit lesson text or video selections; those require a developer.
Past papers	Partial	50 papers are loaded in the database, but the screen shows a fixed list of 30. The PDF files are hosted by a third party (stanmorephysics.com).
Quizzes and progress	Screens only	The quiz engine, points, hearts, streaks and leaderboard all display, but quiz results are not saved and the dashboard shows sample figures. Questions are multiple-choice only.
Premium tier	Not started	The application distinguishes free from premium users, but there is no way to become a premium user. No payment integration exists.
Email	Not started	No welcome, password-reset or certificate emails.
Hosting	Not deployed	There is no staging or production address today.

3.3 An important caveat

> The application has only ever been run in a demonstration mode, without a database, a login service or an AI account. Every feature marked “Built” has been exercised with sample data. Connecting real services is the first step of the remaining work and will surface issues that cannot be found any other way. We will do this on our own development accounts so that it does not depend on you.

4. Work completed this week

The following was completed between 17 and 18 September 2026 and does not depend on any decision from you.

Area	Completed	Effect
Engineering quality	Automated checks now run on every change: code style, type checking, a test suite of 388 tests, and a full production build. Thirteen outstanding code-quality errors corrected. Obsolete deployment files for an approach we are not pursuing removed. Web framework updated to close a published security advisory.	Defects are caught before they reach a student.
Video content	All 28 topics now carry real, verified YouTube videos instead of placeholders: two English videos per topic and one Afrikaans video where a suitable one exists. Each video was checked to confirm it plays inside the application. The two videos previously chosen for Grade 11 Trigonometry had been removed from YouTube by their owner and were replaced.	Every topic page now works. A weekly automated check will flag any video removed in future.

Area	Completed	Effect
Quizzes and progress	Quizzes now load real questions from the database and save every answer. Points, hearts and streaks are calculated on the server and cannot be altered from the student's device. Broken topic links on the Subjects page were repaired and the Past Papers screen now reads from the database.	The core learning loop works end to end.
AI tutor	Moved to the current model generation with a cost-saving cache, a South African day boundary for daily limits, and automatic photo downscaling. Certificates now verify on a readable page rather than raw data. Every call is logged for spend visibility, and a health endpoint was added for monitoring.	Lower cost per hint and a measurable, capped AI bill.
Afrikaans and accessibility	218 interface strings moved into the translation system so the Afrikaans interface is now complete (pending native-speaker review of the new phrases). Password reset was built. Pinch-zoom re-enabled, keyboard skip link, heading structure corrected, and automated accessibility checks added.	An Afrikaans learner sees an Afrikaans application; the product meets basic accessibility expectations.
Security	All nine server routes now verify the caller's identity before acting; previously none did. The administration area now denies access when it cannot confirm a user's role, rather than allowing it. Certificate identifiers are cryptographically random. A database access-policy file has been written for application when the database is provisioned.	Student data will be protected from the first real deployment. Passed two rounds of independent security review; merged.

5. Proposed infrastructure and running costs

5.1 Recommended services

We recommend the following. All prices were checked with the vendors on 17 and 18 September 2026, converted at R16.20 to the US dollar, and exclude VAT. Dollar-priced services will move with the exchange rate.

Requirement	Service	Rationale	Launch cost	Production cost
Application hosting	Vercel	Built by the makers of the web framework we use; deploys directly from the code repository; certificates, content delivery and preview environments included. No server to maintain.	Free tier is limited to non-commercial use; suitable for a closed beta only.	$20 (approx. R325) per month.

Requirement	Service	Rationale	Launch cost	Production cost
Database, login and file storage	Supabase	One service provides all three and the application is already written against it.	Free tier: 500 MB database, 50,000 monthly users. Projects pause after seven days without traffic, so unsuitable for launch.	$25 (approx. R405) per month: 8 GB database, 100,000 monthly users, daily backups.
File storage	Supabase (included)	Very little is stored. Videos remain on YouTube. Certificates are generated on the student's device. Photos sent to the AI tutor are not retained.	None.	None.
AI tutor	Anthropic Claude API	Billed per use. A text hint costs approximately R0.04; a photo scan approximately R0.20. Free users are already limited to five hints and one scan per day.	Pay as you go; a beta of 30 students would cost under R150 per month.	Usage-dependent; see 5.2.
Email	Resend	Welcome, password-reset and certificate emails.	Free: 3,000 emails per month.	Free until roughly 1,500 active students; then $20 (approx. R325) per month.
Domain name	.co.za registrar	Local and recognisable.	Approx. R99 per year.	Approx. R99 per year.
DNS and attack protection	Cloudflare (free plan)	Free enterprise-grade DNS and denial-of-service protection.	None.	None.
Payments (if a paid tier is launched)	PayFast	Credentials already exist. No monthly fee. 3.2% plus R2.00 per card transaction; 2.0% for instant EFT; R8.70 per payout.	None.	Approximately 5% of revenue.
Error monitoring	Sentry (free plan)	Alerts us to failures before students report them.	None.	None at launch volumes.

Summary. A closed beta on free tiers costs approximately R99 per year plus a small AI usage charge. A production deployment with a paid tier costs approximately R730 per month in fixed fees plus AI usage; for a few hundred active students we expect R1,000 to R2,000 per month in total.

The following accounts will need to exist in your name, with your payment details: Vercel, Supabase, Anthropic, Resend, Cloudflare, a domain registrar and PayFast. We will set them up on your behalf.

5.2 AI tutor cost by usage

The AI tutor is the only cost that grows with use. The table assumes every student uses their full free daily allowance on each of 22 school days. Actual use is normally well below this.

Active students	Worst case per month	Likely per month (one quarter of allowance)
50	R440	R110

Active students	Worst case per month	Likely per month (one quarter of allowance)
200	R1,760	R440
500	R4,400	R1,100
1,000	R8,800	R2,200

Premium users are not subject to daily limits. We recommend a monthly spending cap on the AI account, which we will set at provisioning.

5.3 The AWS design

The project files include an earlier design for hosting on Amazon Web Services. Priced as drawn, at Cape Town regional rates, it would cost approximately R4,400 per month before any traffic, of which the high-availability database alone is about R2,000. Reduced to a bare minimum it would still cost approximately R1,300 per month and would require someone on call for infrastructure. It also retains the Supabase dependency for login. We do not recommend it for launch. It remains an option if you have an existing AWS commitment or a data-residency requirement.

6. Decisions required

6.1 Please answer these first

These six decisions change the size of the remaining work by a large factor. They are summarised here and set out in full below.

No.	Decision	Why it matters
1	Minimum launch scope	Grade 12 Mathematics alone is roughly a quarter of the content work.
2	Language priority	Full bilingual coverage is a separate project; English with Afrikaans where available is a small addition.
3	Content ownership	Video review is hours; writing practice questions is the largest remaining content task.
4	Content editing	Determines whether an editing screen is built or content changes go through us.
5	Commercial model	Launching free removes the payments workstream from the launch path.
6	Account ownership	We cannot provision production services until the owner and billing contact are known.

1. **Minimum launch scope.** What is the smallest product you would be happy to launch: all of Grades 10 to 12 in both subjects, a single grade, or a single subject? Is there a fixed date we are working towards?
2. **Language priority.** Is Angelo English-first, Afrikaans-first, or fully bilingual? The interface and the AI tutor already work in both languages. For video, the English ecosystem is roughly one hundred times larger than the Afrikaans one, and the most-watched Afrikaans channel has disabled in-app

playback on almost all of its videos. “English, with Afrikaans where it exists” is a small amount of additional curation; “every topic in both languages” is a separate project. See Appendix B.

3. **Content ownership.** Who will review the video selections and who will write the practice questions? We have loaded provisional video choices for every topic (Appendix C and D), so video review is a few hours of work for someone who knows the syllabus. Practice questions are the larger task and cannot be sourced from YouTube; the question bank currently contains six questions.
4. **Content editing.** Do you need to change videos or topic notes yourself, without a developer? Today only a developer can. A simple editing screen is a modest addition; if sending us a list is acceptable for launch, we will leave it out.
5. **Commercial model at launch.** Will the product launch with a paid tier, or free? No payment code exists. Launching free removes an entire workstream from the launch path.
6. **Accounts.** Who will own the vendor accounts listed in Section 5.1, and who is the billing contact?

6.2 Scope

7. Would you like a private beta with a small group of students before public launch? If so, from which school, and who will recruit them?
8. Is the product Mathematics and Physical Sciences only, or should it be structured to accommodate Mathematical Literacy, Life Sciences or other subjects later?
9. What devices and data conditions are your students on? A video-first curriculum is data-intensive. If most students use prepaid mobile data, we should keep videos short, default to lower quality, and make the written notes useful without pressing play.

6.3 Curriculum content

10. How many practice questions per topic is sufficient, and is multiple-choice acceptable? The engine supports multiple-choice only; NSC papers are largely written-answer.
11. Should questions be adapted from past NSC papers (copyright applies) or written fresh?
12. Which YouTube channels do you trust for CAPS content? Appendix A ranks the channels South African learners actually watch. Kevinmathscience is our suggested default for English.
13. What makes a video suitable: maximum length, South African context, exam technique? We need a rule a reviewer can apply consistently.
14. Is the finished Grade 11 Trigonometry topic the right layout (videos, formula box, worked examples, tips)? We will replicate it across the other 27 topics.
15. Who will sign off that the videos and formulas are correct and match the syllabus? We need a named subject reviewer.
16. Should students earn points for watching a video, and should we record which videos they have watched?
17. Videos are sometimes removed by their owners. We have added a weekly automated check that flags this. Who should receive the alert?
18. The most-watched Afrikaans teacher on YouTube (HouNetOpSukkel, 27,700 subscribers) has disabled in-app playback on almost all their videos. Should we approach them, link out to

YouTube for those videos only, or omit them?

19. The exam countdown shows fixed 2026 NSC dates. Can you confirm them, and who will update them each year?
20. The 50 past-paper PDFs are hosted on stanmorephysics.com. Are you comfortable depending on a third-party site, or should we host the files ourselves, and are we permitted to?

6.4 Premium and payments (only if a paid tier is chosen)

21. What is included in Premium, and at what price? Today the only difference is unlimited AI hints and photo scans.
22. Is PayFast confirmed as the payment provider, is the account live or in test mode, and is the product a monthly subscription, a once-off purchase, or both?
23. Free users receive five AI hints and one photo scan per day. Are these the right limits?

6.5 AI tutor

24. Is the tutor's defined personality (never gives the full answer, replies in 80 words or fewer, uses guiding questions, avoids hollow praise) what you want? Is there anything to add or remove?
25. Should students' conversations with the tutor be stored for your review, or discarded? Storing them helps improve content but adds a privacy obligation.

6.6 Accounts and learner data

26. Students may be under 18. Do you require parental consent at sign-up, and do you have a privacy policy and terms of use we should link to?
27. The leaderboard shows other students' first name, last initial and school. Is that acceptable to you and to the schools?
28. Should sign-in support Google accounts (most learners have one through school), or is email and password sufficient?
29. Who should have administrator access, and do you want a parent or teacher view of a student's progress?

6.7 Brand, hosting and ownership

30. Is "Angelo" the final product name, and do you have a logo? Whose name and logo should appear as the issuer on the AI Skills certificate?
31. Where should the application be hosted and on what domain, and in whose accounts (hosting, DNS and code repository)? See Section 5.
32. Which email address should system emails come from?
33. Is approximately R730 per month in fixed fees plus AI usage an acceptable running budget?

7. Next steps

The following work proceeds on our side without waiting for the decisions above.

Step	What it delivers	Client input needed
Connect real login, database and hosting services on our own development accounts	Every feature tested against real services for the first time	No
Make quizzes save results	Points, streaks, hearts and the progress dashboard become real	No
Move the AI tutor to the current model generation and add a spending cap	Lower cost per hint; no risk of an unbounded bill	No
Complete the Afrikaans interface and add password reset	Afrikaans users see an Afrikaans application; no product ships without password reset	No
Produce a scoped delivery plan with dates	A committed timeline	Yes — decisions 1 to 6

Appendix A. English video channels

The channels below teach the South African CAPS syllabus for Grades 10 to 12 Mathematics and Physical Sciences. Subscriber and view counts were read from YouTube on 17 September 2026.

Rank	Channel	Subscribers	Style	Typical length	Assessment
1	Kevinmathscience	1.11 million; 2,023 videos; 335 million views	One concept or exam question per video, Maths and Physics	2 to 15 minutes	Recommended default. Largest reach by a factor of three, exam-focused, covers every topic. Suited to mobile data budgets.
2	Mlungisi Nkosi Maths and Science	386,000	Full structured lessons, Maths and Physics	20 to 60 minutes	Strongest for Grade 12 Calculus, Sequences and Momentum. Best used as the longer lesson alongside a short clip.
3	Miss Martins Maths and Science	329,000	Classroom lessons and past-paper walkthroughs; Physics and Chemistry as well as Maths	6 to 20 minutes	The best Grade 10 Physical Sciences coverage available.
4	Mindset Learn	302,000; 7,081 videos	Recorded live lessons (Mindset Network, aligned to the Department of Basic Education)	50 to 90 minutes	Official and complete, but long and little-watched today. Useful for topics no one else covers.
5	ThunderEDUC	195,000	Past-paper revision	30 to 60 minutes	Suited to exam-season revision, not first teaching.
6	PHAKAMA RESEARCH	118,000	CAPS-aligned revision, all grades	40 to 60 minutes	Sound but low per-video viewership.
7	Isinamuva Tutoring	55,300	Maths and Science in isiZulu	1 to 2 hours	Worth discussing: the app supports one alternative-language video per slot.
8	Zandile loves Maths	37,800	Long-form “everything you need to know” sessions	2 to 10 hours	Popular, but multi-hour videos are unsuited to an embedded player on mobile data.

Rank	Channel	Subscribers	Style	Typical length	Assessment
9	Wiskunde_Mathematics_SUM	13,600; 926 videos	Short Maths clips with every video in both English and Afrikaans	5 to 15 minutes	The natural choice for the Afrikaans slot; see Appendix B.
10	Simple Science	15,700	Physical Sciences explainers	10 to 20 minutes	Reserve option for physics topics.

Miss Angler (346,000 subscribers) is not listed because the channel covers Life Sciences.

Across all 28 topics, the top search results from these channels were Kevinmathscience 61 times, Miss Martins 28, Mlungisi Nkosi 7 and Mindset 3.

Selection guidance. Two videos per topic are enough to launch: one short introduction and one exam-question walkthrough. Videos under 20 minutes suit mobile data (a 15-minute video is roughly 40 to 60 MB; an hour is 200 MB or more). Watch the first minute: accent, pace and board legibility on a phone matter more than the channel name.

Appendix B. Afrikaans video channels

There is a genuine Afrikaans YouTube ecosystem for CAPS Mathematics and Physical Sciences, but it is roughly one hundredth the size of the English one. The largest channel has 27,700 subscribers, and a typical Afrikaans video has hundreds to low thousands of views rather than hundreds of thousands.

A constraint you should know about. HouNetOpSukkel (27,700 subscribers, 1.87 million views, 63 videos) is by a wide margin the most-watched Afrikaans teacher, with videos averaging 20,000 to 85,000 views. However, 14 of the 16 HouNetOpSukkel videos we found are set by the owner not to play inside other applications; only two Grade 10 Functions videos allow it. This is a setting the channel owner controls and it cannot be overridden. The practical options are to ask the owner to enable it, to send students to YouTube for those videos only, or to use other channels. They are listed in Appendix D marked “cannot embed” so you can see what would be missing.

Rank	Channel	Subscribers	Covers	Assessment
1	HouNetOpSukkel	27,700; 63 videos; 1.87 million views	Wiskunde and Fisiese Wetenskappe, Grades 10 to 12, crash courses and revision	Best content, but in-app playback disabled. Worth one email to the owner.
2	Wiskunde_Mathematics_SUM	13,600; 926 videos; 3.0 million views	Short Maths clips, Grades 8 to 12, every video in both Afrikaans and English	Recommended. Numbered series per topic, short, embeddable, and the same teacher in both languages.
3	Online Maths by Miss Pythagoras	5,920; 1,168 videos; 855,000 views	Afrikaans Maths, Grades 8 to 12, numbered lessons per topic	The most systematic Afrikaans coverage of the Grade 12 syllabus. Embeddable.
4	Wiskunde Broers	4,010; 60 videos	Afrikaans Maths	Small library; one strong Grade 12 proportionality video.
5	Danike Aucamp	2,680; 346 videos	Wiskunde en Wetenskap, Grades 10 to 12, summary format	Useful for the Chemistry side of Physical Sciences. Embeddable.
6	WiskundeMetMargie	1,220; 248 videos	Afrikaans Maths, numbered series, Grades 10 to 11	Best Grade 10 Algebra and Euclidean Geometry coverage in Afrikaans. Embeddable.
7	WSILAB	1,190; 328 videos	Fisiese Wetenskap en Wiskunde, Afrikaans and English	Reserve option for physics.
8	Ekstra Wetenskap Klasse	876	Afrikaans Physics, Grade 10	Small but on-topic: the best Grade 10 circuits and waves clips in Afrikaans. Embeddable.

Rank	Channel	Subscribers	Covers	Assessment
9	Skeinat met Schabort	752; 1,006 videos	Afrikaans Physical Sciences, Grades 10 to 12	Large library, small audience, started November 2024. Did not surface in topic searches; worth a manual look.

Science Clinic (6,250 subscribers) hosts its Afrikaans courses on its own website, not on YouTube, and is therefore not a source for the application.

Coverage. Afrikaans candidates were found for all 28 topics, but no suitable Afrikaans video was found for Grade 10 Mechanics or for Grade 11 wavefronts and diffraction; those topics fall back to English. Several other physics topics have only one or two candidates.

Appendix C. English candidate videos by topic

Search results per topic from the channels in Appendix A, filtered to videos that play inside the application, with view counts read from YouTube on 17 September 2026. Length categories: under 15 minutes is ideal for mobile data; over two hours is not recommended for embedding.

Videos marked ✓ are currently loaded in the application as provisional selections. Reply with the codes of the videos you prefer, or confirm the current selections.

Mathematics · Grade 10 — Algebraic Expressions

Loaded	Code	Title	Channel	Views	Length
✓	EnVbvflrJ4Q	8a) Factorising trinomials grade 10 / Try	Kevinmathscience	344	2:52
✓	52-eRCXTPX0	Grade 10 Maths ALGEBRA Exam Questions Practice	Miss Martins Maths and Science	267	57:07
	6Am2_HXuoqU	Exponential Equation Grade 10	Kevinmathscience	333	1:39
	pirVy1agpus	6) Common Factor / factorisation grade 10	Kevinmathscience	500	5:04

Mathematics · Grade 10 — Equations and Inequalities

Loaded	Code	Title	Channel	Views	Length
✓	OFX7zmbEmro	Grade 10 Inequalities	Kevinmathscience	405	3:15
	LHAq_5VmxRU	Equations Grade 10: Linear Equation without fraction	Kevinmathscience	287	5:16
	FWKCISifl9I	Grade 10 Equations: Linear Equations	Miss Martins Maths and Science	22	6:44
✓	zHNvf3hMlSk	Exam Equations Grade 10	Kevinmathscience	87	4:01

Mathematics · Grade 10 — Functions and Graphs

Loaded	Code	Title	Channel	Views	Length
✓	1RnuVXYcVpE	Exam Functions Grade 10	Kevinmathscience	563	13:37
✓	ZqRXA19yhMM	🚀 Grade 10 Mathematics / Straight Line Functions Made Easy	SmartPrep MathsScience	45	7:30
	Nujt5qcZWqE	Parabola Grade 10 Functions (Introduction)	Miss Martins Maths and Science	229	9:52
	9pJupVoCq7g	Domain and Range for Functions: Grade 10 Maths	Miss Martins Maths and Science	157	21:13

Mathematics · Grade 10 — Number Patterns

Note: Grade 10 linear-pattern videos from the top channels either did not surface or block embedding (My Class Series does). These are Grade 11/12 pattern videos — a manual search is needed for this

topic.

Loaded	Code	Title	Channel	Views	Length
✓	Ws_tyxB-Xyo	Quadratic number patterns grade 11: introduction and examples	Kevinmathscience	475	11:29
✓	GzFLsWU5jlY	Quadratic Pattern Grade 11: Exam	Kevinmathscience	90	9:00
	sXwkDJw8izM	Grade 12 Number Patterns/ Sequences And Series/ Exam Questions	Zandile loves Maths	127	4:55:48

Mathematics · Grade 10 — Finance and Growth

Note: 'Interest Maths Lit' is the Mathematical Literacy version; the other three are Core Maths.

Loaded	Code	Title	Channel	Views	Length
✓	AKU4lSQvrnI	Financial Maths Grade 10 / Simple Interest Introduction	Kevinmathscience	440	3:59
	q9Sxe0HBDR4	Financial Maths Grade 10 / Simple Interest / Determine P	Kevinmathscience	244	2:53
✓	3cj331uG3ul	Financial Maths Grade 10 Past Paper Exam Question 1	Miss Martins Maths and Science	51	8:40
	le6w0YG-ecE	Interest Maths Lit	Kevinmathscience	456	10:17

Mathematics · Grade 10 — Trigonometry

Loaded	Code	Title	Channel	Views	Length
✓	4jSqFXdmKR0	Sin Triangle Grade 10	Kevinmathscience	827	4:23
✓	KvBqPFfzfEM	Trigonometry 2D Grade 10 Angle of elevation and depression	Miss Martins Maths and Science	159	13:49
	Z9ltlFQILHw	Trigonometry Grade 10: Introduction to trig	Miss Martins Maths and Science	156	13:26
	K5Odz79j7to	Grade 10 Trigonometry: Trig Ratios Sin, Cos and Tan	Miss Martins Maths and Science	73	15:39

Mathematics · Grade 10 — Analytical Geometry

Note: The 813K-view Kevinmathscience video is Grade 11 revision of Grade 10 work — fine as the second clip.

Loaded	Code	Title	Channel	Views	Length
✓	VDCxnhnkzJg	Analytical Geometry Grade 10: Distance formula - length of a line	Miss Martins Maths and Science	71	9:59
✓	w1QwFixKjBo	Analytical Geometry Grade 10 Exam Question	Kevinmathscience	440	12:41
	GhiX06fp_rM	Exam Analytical Geometry Grade 10	Kevinmathscience	229	15:12

Loaded	Code	Title	Channel	Views	Length
	QmNzxSM3qZA	gr 11 analytical geometry grade 10 revision	Kevinmathscience	813	14:11

Mathematics · Grade 10 — Euclidean Geometry

Loaded	Code	Title	Channel	Views	Length
✓	lJHk-dG-Or4	Grade 10 Euclidean Geometry Exam Questions Part 1	Miss Martins Maths and Science	190	41:35
	tuAFyRv62wI	Grade 10: Euclidean Geometry / Everything you need to know	Zandile loves Maths	805	3:22:39
✓	g5b1Ygc-Tn4	Exam Questions Grade 10 Euclidean Geometry	Kevinmathscience	236	6:22
	yY9qGF-CjXI	Exam Euclidean Geometry Grade 10	Kevinmathscience	297	7:43

Mathematics · Grade 10 — Statistics

Loaded	Code	Title	Channel	Views	Length
✓	xi8VjMmWdgI	Statistics Grade 10: Mean, mode, median	Kevinmathscience	770	4:45
✓	Hc-BUqBaUkQ	Statistics Grade 10: Exam Question	Kevinmathscience	293	2:50
	oVtvnQ8OW5Y	Statistics Grade 10: Table	Kevinmathscience	439	6:59
	94_ljD-J19Y	Statistics Grade 10: Box and whisker	Kevinmathscience	471	4:46

Mathematics · Grade 10 — Probability

Loaded	Code	Title	Channel	Views	Length
✓	Hqbbyh5OEtc	Probability Grade 10 / Introduction	Kevinmathscience	630	6:25
	vgAf7PxSK7E	Probability Grade 10 / Tree Diagram	Kevinmathscience	200	8:48
✓	iBoBTRrp-w8	Probability Grade 10 / Exam Question	Kevinmathscience	350	5:51
✓	kMJVyxYiato	Probability Maths Lit Part 1	Kevinmathscience	390	8:58

Mathematics · Grade 11 — Trigonometry

Loaded	Code	Title	Channel	Views	Length
✓	IhXnVk8PuJY	Trigonometry Grade 11 Exam Questions	Kevinmathscience	225	4:45
✓	Optfjw6HGZw	Trig functions grade 11 and 12: Determine Equation	Kevinmathscience	434	3:27
✓	rEFv51neoUc	General Solutions Grade 11 Trigonometry PART 1	Miss Martins Maths and Science	4	14:26
✓	cZoEfccMapk	Trig Functions Grade 11 Exam Questions	Kevinmathscience	14	9:56

Mathematics · Grade 11 — Analytical Geometry

Loaded	Code	Title	Channel	Views	Length
✓	VB4jH92V-tg	Grade 11 Analytical geometry: angle of inclination	Kevinmathscience	563	6:49
	mLAhC1DHtDc	Grade 11 analytical geometry: 4th Co-ordinate of parallelogram	Kevinmathscience	455	2:55
	nvHOzaZKxbA	Grade 11 Analytical Geometry Grade 10 Revision	Miss Martins Maths and Science	21	19:48
✓	lrmASs4NOww	Grade 11 Analytical Geometry Angle of Inclination Introduction and practice	Miss Martins Maths and Science	23	12:25

Mathematics · Grade 11 — Euclidean Geometry

Loaded	Code	Title	Channel	Views	Length
	4NqwU3BzOcE	Grade 11: Euclidean Geometry / Everything you need to know	Zandile loves Maths	670	5:45:01
✓	bc6x2GQcMbU	Circle Geometry Grade 11 : Practice	Kevinmathscience	151	12:14
✓	lQE642ILPDY	Euclidean Geometry Grade 11: Exam question	Kevinmathscience	412	20:18
	KWYgy_YkyCA	Grade 11 - Euclidean Geometry / Tangents and chords / Mlungisi Nkosi	Mlungisi Nkosi Maths & Science	30	22:02

Mathematics · Grade 11 — Statistics

Loaded	Code	Title	Channel	Views	Length
✓	CTj0OE6lkkg	Statistics Grade 11: Ogive curve	Kevinmathscience	832	6:45
	aNv4gD72-1I	Statistics Grade 11	Kevinmathscience	12	13:11
	ifJUTSTL4WU	Statistics Grade 11: Table	Kevinmathscience	738	4:08
✓	Amf60jAYIaI	Statistics Grade 11 Exam Questions	Kevinmathscience	222	8:32

Mathematics · Grade 12 — Calculus

Loaded	Code	Title	Channel	Views	Length
✓	UzVKkR-XArM	Determine Cubic Equation Calculus grade 12	Kevinmathscience	469	4:28
	4VUWIiudCk0	Limits Calculus Grade 12	Kevinmathscience	10	5:08
✓	1ye4FhvMazo	Lesson 7: How To Solve Optimisation Questions / Area, Volume & Cost / Grade 12 Calculus	Mlungisi Nkosi Maths & Science	286	21:55
	_WFx3JDTuQg	Lesson 4: How To Plot and Interpret Cubic Functions / Grade 12 Calculus	Mlungisi Nkosi Maths & Science	249	51:37

Mathematics · Grade 12 — Sequences and Series

Loaded	Code	Title	Channel	Views	Length
	An_O1nY1lac	17) Worksheet Series Sequences / Half	Kevinmathscience	228	1:53:07
	S9oGEqRqO_8	Grade 12 Algebra Lesson 1/ Sequences and Series : Arithmetic Sequence Explained	Mlungisi Nkosi Maths & Science	400	24:34
✓	nBFLxEZko8o	Grade 12 Mathematics / Sequence and Series / Past Exam Questions / May/June 2021_Question 2	Mlungisi Nkosi Maths & Science	112	23:09
✓	DlOaZtU7XlA	Exam Question Series Sequences	Kevinmathscience	79	7:07

Mathematics · Grade 12 — Analytical Geometry

Loaded	Code	Title	Channel	Views	Length
	H4m6YuW75vw	Grade 12: Analytical Geometry / Everything you need to know	Zandile loves Maths	97	2:03:08
✓	O9mO17v2-e8	Analytical Geometry Grade 12 Exam Questions	Kevinmathscience	190	20:25
✓	LxMklbspJAw	Analytical Geometry Grade 12 Exam Questions	Kevinmathscience	97	12:29
	RU1Uott6Xf4	Analytical Geometry Grade 12 Exam Question	Kevinmathscience	316	12:28

Mathematics · Grade 12 — Euclidean Geometry

Loaded	Code	Title	Channel	Views	Length
	9RJLHVysqA8	Grade 12: Euclidean Geometry / Everything you need to know	Zandile loves Maths	140	10:45:13
	kwsSjMvqTF4	Grade 12: Prop Theorem and Similarity Updated / Euclidean Geometry / Everything you need to know	Zandile loves Maths	2	5:27:45
✓	nWn-HjCP9wY	Grade 12 Euclidean Geometry Exam question # 1	Kevinmathscience	255	6:26
✓	122eVl0o_C8	EUCLIDEAN GEOMETRY / PROPORTIONALITY GRADE 12 [ISIZULU]	Isinamuva Tutoring	73	1:48:45

Mathematics · Grade 12 — Probability

Note: 'Probability Maths Lit Part 1' is Mathematical Literacy, not Core Maths.

Loaded	Code	Title	Channel	Views	Length
	1IpyhSJvKiY	Grade 12: Probability / Everything you need to know	Zandile loves Maths	94	2:31:34
✓	kMJVyxYiato	Probability Maths Lit Part 1	Kevinmathscience	390	8:58

Loaded	Code	Title	Channel	Views	Length
✓	loYKqFQcksY	Gr 12 probability # 1	Kevinmathscience	675	3:48
	0haRHA-QoAk	PROBABILITY GRADE 12 / COUNTING PRINCIPLES [ISIZULU]	Isinamuva Tutoring	39	58:08

Physical Sciences · Grade 10 — Mechanics

Loaded	Code	Title	Channel	Views	Length
✓	gtiOPVVKkU4	Grade 10 Mechanics Past Paper Question Motion in One Dimension	Miss Martins Maths and Science	48	10:00
	vW93Ub-vsD4	Motion in one dimension: Frame of reference and position Grade 10 Physics	Miss Martins Maths and Science	107	8:39
✓	-wHXFMMEltw	Grade 10 Mechanics: Equations of Motion Exam Question PART 1	Miss Martins Maths and Science	40	6:06
	7-vQ6k4bTqM	Equations of Motion Grade 10	Kevinmathscience	264	22:32

Physical Sciences · Grade 10 — Transverse Waves

Loaded	Code	Title	Channel	Views	Length
	JUnMAK9qPNU	3) Transverse Waves Grade 10	Kevinmathscience	388	24:09
	4vDk3W66s5I	Grade 10 Physics Transverse waves Period and Frequency	Miss Martins Maths and Science	100	15:05
✓	PGgeSD7vPOQ	Grade 10 Transverse Waves Points in phase and points out of phase	Miss Martins Maths and Science	81	9:28
✓	o7BDo4CcQOI	Grade 10 Transverse waves Exam question Wavelength	Miss Martins Maths and Science	99	13:10

Physical Sciences · Grade 10 — Sound

Loaded	Code	Title	Channel	Views	Length
✓	1u5GSxuSj7A	5) Sound Waves Grade 10	Kevinmathscience	214	19:55
✓	4IpqrR4vqOw	Grade 10 Physics Sound waves	Miss Martins Maths and Science	86	11:22
	9sP74zJKcIQ	4) Longitudinal Waves Grade 10	Kevinmathscience	207	5:59
	JUnMAK9qPNU	3) Transverse Waves Grade 10	Kevinmathscience	388	24:09

Physical Sciences · Grade 10 — Electromagnetic Radiation

Loaded	Code	Title	Channel	Views	Length
✓	a4aHyWtV2Kw	Grade 10 Electromagnetic Radiation Physics: Introduction	Miss Martins Maths and Science	228	13:22

Loaded	Code	Title	Channel	Views	Length
✓	AAyPp8W9mWw	Electromagnetic radiation calculations Grade 10 Physics	Miss Martins Maths and Science	196	12:40
	b1JrapSyaIQ	Grade 10 Physics Particle Nature of Electromagnetic Radiation	Miss Martins Maths and Science	87	8:02
	18mpmoo4g7A	Electrostatics Grade 10 / Introduction	Kevinmathscience	347	25:25

Physical Sciences · Grade 10 — Electric Circuits

Loaded	Code	Title	Channel	Views	Length
✓	imoShDLnQqE	Circuits grade 10 / Part 1	Kevinmathscience	348	10:13
✓	5WRqdTr9WmQ	Circuits Grade 10 / Calculations	Kevinmathscience	295	29:44
	hMAArBq00y4	Electricity Grade 10 Electric Circuits Summary Physics	Miss Martins Maths and Science	179	58:00
	JZJlJ18ggc0	Resistance Grade 10	Kevinmathscience	212	12:33

Physical Sciences · Grade 10 — Matter and Materials

Loaded	Code	Title	Channel	Views	Length
✓	5M4hJjp3-lQ	Grade 10 Classification of Matter: Properties of materials	Miss Martins Maths and Science	118	9:46
✓	oADmMF84wbA	Water of Crystallization / Chemistry Grade 10-12	Miss Martins Maths and Science	272	17:52
	B5gVS4Ti3P8	Cooling curve vs Heating curve Grade 10 Chemistry	Miss Martins Maths and Science	75	10:43
	SSKvnWYbrwM	Grade 10 Matter and Materials Homogeneous and Heterogeneous mixtures	Miss Martins Maths and Science	43	11:37

Physical Sciences · Grade 11 — Newton’s Laws

Loaded	Code	Title	Channel	Views	Length
✓	AFwr05QW3aY	Newton Law Gravity Grade 11	Kevinmathscience	255	19:38
	aSC1sgTUODA	Newton’s second law Grade 11 and 12 [IsiZulu]	Isinamuva Tutoring	71	1:01:49
✓	HmoNMxId4-Y	Grade 11 Newton Laws: Objects on a slope	Kevinmathscience	552	7:47
	7YhkjSlxsrc	Grade 11 Newton Laws: Look out for normal force	Kevinmathscience	504	7:17

Physical Sciences · Grade 11 — Waves and Sound

Note: Under CAPS, waves and sound are Grade 10 work; Grade 11 covers 2D/3D wavefronts and diffraction. The app's topic list may be mislabelled — the subject reviewer should confirm. Candidates below are the Grade 11 wavefront content.

Loaded	Code	Title	Channel	Views	Length
✓	IQ_HH68wmHw	2D & 3D Wavefronts	Mindset	4	1:14:58
✓	KXRnmQGweJE	Grade 11 - Diffraction / Huygen's Principle & Wavefronts / November 2017 / Mlungisi Nkosi	Mlungisi Nkosi Maths & Science	333	4:52
	lf689C99Sxw	Diffraction	Mindset	2	50:49
	5–qGnkJtN4	Revision: Refraction & Diffraction	Mindset	10	53:36

Physical Sciences · Grade 12 — Momentum and Impulse

Loaded	Code	Title	Channel	Views	Length
	QAURZq43508	Physics / Momentum / Conservation of linear momentum (Made easy)	Mlungisi Nkosi Maths & Science	388	1:02:20
	xfTAxHUR1pU	PHYSICAL SCIENCES GRADE 12 - MOMENTUM AND IMPULSE REVISION MODERATED	PHAKAMA RESEARCH	1	45:12
✓	JOyL0KJRe0g	1) Momentum Gr 12 / Intro	Kevinmathscience	770	2:09
✓	7Zq7PjYaySU	Exam question momentum gr 12	Kevinmathscience	208	17:17

Appendix D. Afrikaans candidate videos by topic

Afrikaans search results per topic, curated to remove off-topic results. Videos marked “cannot embed” are set by their owner not to play inside other applications (see Appendix B).

Videos marked ✓ are currently loaded in the application as provisional selections. Reply with the codes of the videos you prefer, or confirm the current selections.

Wiskunde (Mathematics) · Graad 10 — Algebraic Expressions

Loaded	Code	Title	Channel	Views	Length
✓	68tBHlCty1M	Graad 10 Algebraïese Uitdrukkings #1: Die Getallestelsel	WiskundeMetMargie	27	05:38
	xl0AdY6jZUs	Graad 10 Algebraïese Uitdrukkings #6: Produkte met Tweeterms en Drieterms	WiskundeMetMargie	15	12:25
	NNyGntxiNzk	Graad 10 Algebraïese Uitdrukkings #7: Faktorisering (Kruismetode en Vele Meer)	WiskundeMetMargie	14	13:47
	de4BWka2t5g	Gr 10 Wiskunde : Faktorisering - Som en Verskil van Derdemagte (cannot embed)	HouNetOpSukkel	256	10:18

Wiskunde (Mathematics) · Graad 10 — Equations and Inequalities

Note: The three strong videos are all HouNetOpSukkel and cannot be embedded. The only embeddable one has 45 views. Weak topic.

Loaded	Code	Title	Channel	Views	Length
	tU4eNI2SwAE	Gr 10 Wiskunde : Gelyktydige Vergelykings - Substitusie en Eliminasie (cannot embed)	HouNetOpSukkel	406	25:26
	tiWXFqpqsgs	Gr 10 Wiskunde : Lineêre Ongelykhede (cannot embed)	HouNetOpSukkel	267	18:56
	QcmyvFc1r9U	Gr 10 Wisk : Hersieningsklas 2.1 - Vergelykings en Ongelykhede (cannot embed)	HouNetOpSukkel	145	20:06
✓	8IUh8jx2vrM	Gr10 (CAPS) Vergelykings en Ongelykhede	Philpotts Consultancy	45	10:09

Wiskunde (Mathematics) · Graad 10 — Functions and Graphs

Loaded	Code	Title	Channel	Views	Length
	oORnjRomY5w	Gr 10 Wiskunde : Funksies - Parabool Intro	HouNetOpSukkel	476	10:04
	Fv-g8e-ZZdE	Gr 10 Wiskunde : Funksies - Bepaal die Vergelyking van ’n Parabool	HouNetOpSukkel	310	11:13
✓	90DcF3f3V7c	Funksies Parabool Graad10: L6 Grafiek Interpretasie Eksamen vraag	Online Maths by Miss Pythagoras	293	08:11

Loaded	Code	Title	Channel	Views	Length
	i4bKrt8Puk4	Skets Van Funksies En Grafieke // Wiskunde Graad 10	CollieMath Studio	532	11:26

Wiskunde (Mathematics) · Graad 10 — Number Patterns

Loaded	Code	Title	Channel	Views	Length
	5TV5cYTMytA	Graad 10 Wiskunde: Linieere Getalpatrone	Dalene Coetzee	898	11:30
	0y9fAfDp3TM	Getalpatrone Graad10: L1 Hersiening en Inleiding	Online Maths by Miss Pythagoras	569	06:01
	jAMB56RrfDs	W10 Getalpatrone graad 10	Stefan Buys	22	08:27
✓	C7igJjQ_rsM	Graad 10 Wiskunde Hersiening: Getalpatrone / Vrae uit Vorige Vraestelle (KABV)	Smart Culture Education	625	09:39

Wiskunde (Mathematics) · Graad 10 — Finance and Growth

Note: Last video is Grade 11 level. Mathematical Literacy results were removed.

Loaded	Code	Title	Channel	Views	Length
✓	T7zTTrn1ulE	Graad 10 Wiskunde Les 1 Enkelvoudige en saamgestelde rente 13 tot 17 Julie	Rochelle Van der Merwe	18	08:14
	K08zwu3OID0	Graad 10 / Verstaan Enkelvoudige Rente: 'n Inleiding	Smart Culture Education	495	03:55
	mGla5BYX9vM	Enkelvoudige en Saamgestelde Rente - Graad 11	SlimDink	21	08:51

Wiskunde (Mathematics) · Graad 10 — Trigonometry

Loaded	Code	Title	Channel	Views	Length
✓	U3rGPy484Ro	Graad 10 Trigonometrie	YCDM - YouCanDoMaths	128	08:56
	aYs5h0288mg	Gr10 Trigonometrie - Les 5 Sakrekenaar Spesiale hoeke	Lekker Wiskunde	289	11:49
	pNasYZnXYSw	Gr. 10 Wiskunde: Trigonometrie	Danike Aucamp_Wiskunde en Wetenskap	678	28:06
	ZVc-YFUfwtE	Gr 10 Wiskunde : Trigonometrie - Oplos van Driehoeke (cannot embed)	HouNetOpSukkel	846	36:44

Wiskunde (Mathematics) · Graad 10 — Analytical Geometry

Loaded	Code	Title	Channel	Views	Length
	jndysNtUg9s	Analitiese meetkunde Middelpunt Graad 10 en ouer (wiskvaw)	VAW-Vereniging vir Afrikaanse Wiskunde-onderwysers	22	02:12
	arh51glLHFU	W10 3.Analitiese Meetkunde Hellings/Gradiënte	Stefan Buys	419	04:46
✓	4kCOeJdhECM	Gr10 Junie Eksamen voorbereiding: Analitiese Meetkunde	Lichelle Barnard	116	24:15
	0EgRpXmKq44	Gr 10 Wisk : Hersieningsklas 4 - Analitiese Meetkunde (cannot embed)	HouNetOpSukkel	500	12:53

Wiskunde (Mathematics) · Graad 10 — Euclidean Geometry

Loaded	Code	Title	Channel	Views	Length
	8ev_al-joQI	Graad 10 Euklidiese Meetkunde #1: Inleiding	WiskundeMetMargie	25	08:06
	37yh72NntxQ	Graad 10 Euklidiese Meetkunde #10 : Middelpuntstelling met Voorbeelde	WiskundeMetMargie	608	16:39
	oda9jTEWVyU	Graad 10 Euklidiese meetkunde Parallelo gramme	Wiskunde	100	14:10
✓	BSDa-8njr2w	Euklidiese Meetkunde Voorbeelde - Verduideliking // Wiskunde Graad 10	CollieMath Studio	382	01:09:43

Wiskunde (Mathematics) · Graad 10 — Statistics

Loaded	Code	Title	Channel	Views	Length
✓	twoRWkbvv_o	Graad 10 Wiskunde Les 1 Statistiek	Rochelle Van der Merwe	70	04:12
	IyFhkpXp9zM	Gr. 10 - Statistiek Video 1	QuickMafs	18	04:23
	GuM1joF5aWw	Graad 10 Wiskunde Memo 1 Statistiek	Rochelle Van der Merwe	377	05:28
	aYxCSk3zeTg	Data Hantering Voorbeeld Verduideliking // Wiskunde Graad 10	CollieMath Studio	893	24:48

Wiskunde (Mathematics) · Graad 10 — Probability

Loaded	Code	Title	Channel	Views	Length
	Gx2qBMBObYs	01 Waarskynlikheid - Hersiening van Graad 10	Wiskunde_Mathematics_SUM	196	12:13
✓	ZRZQV5YLuPc	Graad 10 Waarskynlikheid	YCDM - YouCanDoMaths	130	13:45
	ZXDCP67cYEw	Graad 10 Wiskunde Les 1 Waarskynlikheid	Rochelle Van der Merwe	33	07:13

Loaded	Code	Title	Channel	Views	Length
	pLP_eWbWoVk	Graad 10 Wiskunde Les 4 Waarskynlikheid	Rochelle Van der Merwe	704	01:57

Wiskunde (Mathematics) · Graad 11 — Trigonometry

Loaded	Code	Title	Channel	Views	Length
✓	Jiz_6sQY418	Graad 11 Trig identiteite (bewys)	Mathfanatics	452	18:46
	WLeO-VJj-zs	Voorbeeld van Bewys die Identiteit (LK=RK) - Graad 11	SlimDink	818	03:59
	aZ31zEGOhgs	Herkende Trigonometriese Identiteite - Graad 11	SlimDink	138	08:28
	ImrMU3AKWr8	Trigonometrie ; Bewys Van Identiteite - Verduideliking en Voorbeelde // Wiskunde Graad 11	CollieMath Studio	80	29:50

Wiskunde (Mathematics) · Graad 11 — Analytical Geometry

Note: Only one on-topic embeddable Afrikaans video surfaced. Weak topic.

Loaded	Code	Title	Channel	Views	Length
✓	kwLfg5jJ2eI	Graad 11 Analitiese Meetkunde #6: Ewewydige (Parallele) Lyne	WiskundeMetMargie	181	12:47
	0EgRpXmKq44	Gr 10 Wisk : Hersieningsklas 4 - Analitiese Meetkunde (cannot embed)	HouNetOpSukkel	500	12:53

Wiskunde (Mathematics) · Graad 11 — Euclidean Geometry

Loaded	Code	Title	Channel	Views	Length
✓	ULsEMok0Ksk	Sirkelmeetkunde vraagstukke 1 euklidiese graad 11 (wiskvaw) (wiskdo)	VAW-Vereniging vir Afrikaanse Wiskunde-onderwysers	915	04:42
	cCej2oFSZ1s	Sirkelmeetkunde vraagstukke 3 euklidiese graad 11 (wiskvaw) (wiskdo)	VAW-Vereniging vir Afrikaanse Wiskunde-onderwysers	229	04:53
	iU6JUwI7l8w	Grade 11 euklidiese meetkunde sirkels (mathdou)	Dou Kleefstra	21	04:49
	HBw5Lu95IbM	Euklidiese meetkunde sirkels 11.20 (wiskdou) Graad 11 en ouer	Dou Kleefstra	23	02:57

Wiskunde (Mathematics) · Graad 11 — Statistics

Loaded	Code	Title	Channel	Views	Length
	FAkpy9ll958	04 Gegroepeerde data en ogiewe	Wiskunde_Mathematics_SUM	138	09:49
	w_-oF0w5DDY	Statistiek Graad11 L2: Ogief of kumulatiewe frekwensie kurwe Wiskunde	Online Maths by Miss Pythagoras	50	10:29
✓	v0NWn7nXALU	Gr 11 en 12 Ogief Statistiek Deel 1 - Die Basics	But They Ask It In A Funny Way	178	15:28

Wiskunde (Mathematics) · Graad 12 — Calculus

Loaded	Code	Title	Channel	Views	Length
	7xLX-oLsmFM	09 Grafiese interpretasie van afgeleide	Wiskunde_Mathematics_SUM	43	05:47
	vgXpuSdQbjc	Die grafiek van die afgeleide van ’n funksie	Wiskunde Graad 12 Juffrou Koorsen	82	08:22
	T7IN-6kcGE0	Graad 12 Differensiaalrekene Kort metode	Wiskunde	23	25:55
✓	n3XE5NtY-Ls	Differensiasie - Verduideliking en Voorbeelde // Wiskunde Graad 12	CollieMath Studio	341	36:12

Wiskunde (Mathematics) · Graad 12 — Sequences and Series

Note: HouNetOpSukkel has the best three (20–30K views each) but all block embedding. One embeddable alternative with 362 views.

Loaded	Code	Title	Channel	Views	Length
✓	Pu2B6FY7YkA	Graad 12, Vraestel 1, Vraag 2, Rye en Reekse Video 1: Teorie	Wizzkunde	362	08:16
	EGLMIlomhjM	Gr 12 Wiskunde : Rye en Reekse - Sigma Notasie (cannot embed)	HouNetOpSukkel	298	24:07
	FZFIkpQgD8s	Gr 12 Wiskunde : Rye en Reekse - Bewys Sn vir ’n Meetkundige Reeks (cannot embed)	HouNetOpSukkel	200	05:34
	cxg1zOYHBng	Gr 12 Wiskunde : Rye en Reekse - Bewys Sn vir ’n Rekenkundige Reeks (cannot embed)	HouNetOpSukkel	236	03:28

Wiskunde (Mathematics) · Graad 12 — Analytical Geometry

Loaded	Code	Title	Channel	Views	Length
✓	9YT8XPAM9Vc	Analitiese Meetkunde Graad 12 Wiskunde: L1 Sirkel met oorsprong middelpunt	Online Maths by Miss Pythagoras	68	13:48

Loaded	Code	Title	Channel	Views	Length
	w5nJdK5AdFQ	04 Raaklyne aan sirkels	Wiskunde_Mathematics_SUM	42	09:07
	v3HhBGG552c	A1 Gr 12 Analitiese Meetkunde Sirkel by oorsprong	Wiskunde Graad 12 Juffrou Koorsen	773	05:03
	Pwf6Ukzv38I	Gr12 Analitiese Meetkunde : Sirkels Notas - Snyding (Interne en Eksterne)	123 Maths Training	91	03:12

Wiskunde (Mathematics) · Graad 12 — Euclidean Geometry

Loaded	Code	Title	Channel	Views	Length
	1dqCThLt2Fk	Euklidiese Meetkunde Graad12: L1 Ewerediqheid in Driehoeke(Lyn ewewydig een sy van driehoek)	Online Maths by Miss Pythagoras	79	15:40
	Sj5pFbuGtK4	Ewerediqheid	Wiskunde Broers	77	08:19
	ffTCN19mHMw	Euklidiese Meetkunde Gemengde Maklike Voorbeelde #1 - #7 // Wiskunde Graad 12	CollieMath Studio	155	49:03
✓	GbXpQ2YDdNQ	Ewerediqheid en Gelykvormigheid Verduideliking Via Voorbeelde // Wiskunde Graad 12	CollieMath Studio	247	01:21:17

Wiskunde (Mathematics) · Graad 12 — Probability

Loaded	Code	Title	Channel	Views	Length
✓	5HNopMSeyLg	Waarskynlikheid Graad12: L1 Fundamentele Telbeginsel Wiskunde	Online Maths by Miss Pythagoras	59	10:56
	kOSiMdRRPBg	Telbeginsels Waarskynlikheid Graad 12 (wiskdou)	Dou Kleefstra	32	04:49
	pnZ3qXnhT08	Waarskynlikheid - Fundamentele Telbeginsel	Arnold Lamont	26	09:02
	0gtdb64d5jg	Wiskunde met LitNet: graad 12 – Waarskynlikheid	Chris Brunette	221	12:23

Fisiese Wetenskappe (Physical Sciences) · Graad 10 — Mechanics

Note: No on-topic Afrikaans video surfaced for motion in one dimension. Search returned wave videos. Needs a manual search or an English fallback.

Loaded	Code	Title	Channel	Views	Length

Fisiese Wetenskappe (Physical Sciences) · Graad 10 — Transverse Waves

Loaded	Code	Title	Channel	Views	Length
✓	fhCoYWh97Nc	Transversale golwe: Graad 10 Fisika	Ekstra Wetenskap Klasse	34	06:46

Loaded	Code	Title	Channel	Views	Length
	IoqhxiZgT5U	Gr 10 Fisiese Wetenskappe Transversale Golwe - Janiena Daubert	Hoërskool Noordheuwel	29	54:19
	tiXt-JRl2tQ	Gr 10 Fisika : Hersieningsklas 1 - Transversale en Longitudinale Golwe (cannot embed)	HouNetOpSukkel	381	21:17

Fisiese Wetenskappe (Physical Sciences) · Graad 10 — Sound

Loaded	Code	Title	Channel	Views	Length
	9NSTY53rU0o	Gr.10 Longitudinale golwe	Douwliena Fourie	741	03:31
✓	Ye7QmYuHeBE	Gr 10 Fisiese Wetenskappe Longitudinale Golwe - Janiena Daubert	Hoërskool Noordheuwel	19	23:13
	tiXt-JRl2tQ	Gr 10 Fisika : Hersieningsklas 1 - Transversale en Longitudinale Golwe (cannot embed)	HouNetOpSukkel	381	21:17

Fisiese Wetenskappe (Physical Sciences) · Graad 10 — Electromagnetic Radiation

Loaded	Code	Title	Channel	Views	Length
	yVipv9v9JVo	Elektromagnetiese golwe en spektrum Graad 10 Fisika	Ekstra Wetenskap Klasse	22	09:55
✓	nxug2Fhnud8	Gr.10 Formuleblad. Elektromagnetiese golwe.	Douwliena Fourie	435	01:20
	cXkvDhvqQO0	Gr 10 Fisika : Hersieningsklas 2 - Elektromagnetiese Golwe (cannot embed)	HouNetOpSukkel	201	10:39

Fisiese Wetenskappe (Physical Sciences) · Graad 10 — Electric Circuits

Loaded	Code	Title	Channel	Views	Length
✓	xO7N97B1RB0	Stroombane: Berekeninge Graad 10 Fisika	Ekstra Wetenskap Klasse	47	07:08
	fc9fd9d5J14	Stroombane: Voorbeelde van stroombaan berekeninge Graad 10 Fisika	Ekstra Wetenskap Klasse	90	10:45
	efdBSXXC85A	Gr 10 Fisika : Hersieningsklas 4 - Stroombane 1 (cannot embed)	HouNetOpSukkel	402	17:40
	fe58DwqVx9o	Gr 10 Fisika : Hersieningsklas 4 - Stroombane 2 (cannot embed)	HouNetOpSukkel	193	13:10

Fisiese Wetenskappe (Physical Sciences) · Graad 10 — Matter and Materials

Note: Thin: one general chemistry overview and one long lab video.

Loaded	Code	Title	Channel	Views	Length
✓	_Ad0WdFmEsY	Gr. 10 Chemie: Fisiese en Chemiese verandering in ’n NEUTEDOP	Danike Aucamp_Wiskunde en Wetenskap	44	12:58
	69ybb2h55aU	Gr 10 Fisiese Wetenskappe: Erken Anione in oplossings	Mari Bartholomae	32	39:31

Fisiese Wetenskappe (Physical Sciences) · Graad 11 — Newton’s Laws

Loaded	Code	Title	Channel	Views	Length
✓	FA5DBBNyS8E	Newton se wette: Trek krag teen ’n hoek	HenBot Wetenskap	45	13:46
	ZT1bhZDYBD8	Fisika / Newton se Wette Eksamenvraag 6 / Fisiese Wetenskappe CAPS / Gr 11 - 12	Shap Science SA - Juffrou R	110	27:27
	8nD6e9mqQBY	Gr 11 Wetenskap : Meganika - Newton 2 Vb 1 (cannot embed)	HouNetOpSukkel	499	07:26
	jPxiSarJxE4	Gr 11 Wetenskap : Meganika - Newton 2 Vb 4 (cannot embed)	HouNetOpSukkel	196	09:28

Fisiese Wetenskappe (Physical Sciences) · Graad 11 — Waves and Sound

Note: No on-topic Afrikaans video surfaced for 2D/3D wavefronts or diffraction. Results were vectors and redox. Use the English Mindset videos.

Loaded	Code	Title	Channel	Views	Length

Fisiese Wetenskappe (Physical Sciences) · Graad 12 — Momentum and Impulse

Loaded	Code	Title	Channel	Views	Length
✓	2YERVz7HUMI	Momentum en Impuls - Afrikaans Potgooi / Fisika / Fisiese Wetenskappe / Graad 12 NSS	Simple Science	228	08:20
	q7-CZbl6M8k	Boost Series: Momentum en Impuls	STEM in ACTION MandelaUni	835	41:03
	TRrTMvHCh08	Newton se Tweede Wet en Momentum / Graad 12 Fisiese Wetenskap / CAPS Afrikaans	Nelius Maths and Science Tutor	4	21:39
	9x–nIiaIH8	Gr 12 Nas : Meganika - Momentum en Impuls (cannot embed)	HouNetOpSukkel	311	24:54